Estimating the residual differential OPD of a dual heterodyne interferometer from frequency-modulated phasemeter data

Measurement principle, the leakage-immune demodulation pipeline, and three test cases from the FM1 campaign

HET IFO OPD analysis

July 1, 2026

Abstract

A known single-tone modulation of the laser frequency turns a heterodyne interferometer pair into a self-calibrating sensor of its residual optical path-length difference (OPD): the modulation writes a phase tone on the differential interferometer phase whose amplitude is proportional to the OPD, independently of the laser wavelength and of the large, slowly drifting absolute interferometer phase. This note derives that relation, describes the FM1 unit and its two dual-interferometer systems, explains why the differential phase is the right observable (it rejects the common laser frequency noise by $\sim 606\times$ in this data), and documents the estimation pipeline implemented in `het_ifo_opd`: a complex-baseband demodulator that is immune to the spectral leakage which biases a naive lock-in, an automatic coherent/incoherent integration choice driven by a measured tone coherence, and a statistically calibrated uncertainty equal to the Cramér-Rao bound. We then put the pipeline to the test on three deliberately different FM1 acquisitions: one anchored and clean (system R1), and two released acquisitions from the two systems R1 and R2, one strongly leakage-biased and one genuinely phase-incoherent. From these we draw out the pitfalls that dominate this measurement in practice: spectral leakage from the low-frequency wander, real test-mass motion masquerading as measurement scatter, actuator nonlinearity, and calibration of the frequency deviation. All numbers and figures in this note are regenerated from the raw data by `make_figures.py` (`make`).

Contents

1 Scope: the instrument and the measurement problem

1.1 The FM1 unit and its two dual-interferometer systems

The FM1 unit comprises two nominally identical dual-interferometer systems, labelled R1 and R2. Each system is a pair of heterodyne interferometers fed by a common laser: a reference interferometer with static arms, and a measurement interferometer one of whose arms terminates on a test mass. The two interferometers of a system have nearly, but not exactly, matched optical path lengths, so their difference is a small residual differential OPD. Each interferometer is read out on one phasemeter channel, and the analysis works with the difference of the two channels.

The test mass can be held in two states. When it is anchored it is clamped so it cannot move, and the residual differential OPD is static; measuring it well is then a pure precision problem. When the test mass is released it is free to move, and its displacement enters the measurement interferometer directly, so the differential OPD becomes a time-varying, high-precision probe of the forces acting on the mass. A released acquisition therefore has a differential OPD that is dynamic by construction, a point that is central to interpreting the results below.

The acquisition file names encode, in order, the day, the test-mass state (anchored or released), the medium (air or vacuum), the mounting configuration, and the system (R1 or R2). Two mounting descriptors appear: whether the unit is shimmed, and whether flight torque has been applied at its mechanical interface. Both describe only how the FM1 unit is mounted to its interface and are expected to have little or no effect on the differential OPD; confirming that expectation is one aim of the campaign.

1.2 The measurement problem

The difficulty is one of dynamic range. The absolute OPD of each interferometer is of order tens of millimetres (roughly 45 to 70 mm across these systems), and the laser frequency noise coupling through it drives the raw per-channel phase through tens of cycles over a 600 s record. The residual differential OPD is smaller than the absolute OPD by one to two orders of magnitude, and, for an anchored system, the modulation tone that encodes it is smaller still: of order 10^{-4} cycles, four to five decades below the raw phase excursion. Extracting it reliably is as much a signal-processing problem as a physics one.

This note has two purposes. First, to record the measurement principle and the estimator in enough detail that the numbers can be trusted and reproduced (Sections 2 to 3). Second, to exercise the pipeline on three real acquisitions that span the regimes it must cope with, and to distil from them the practical pitfalls to avoid when taking these measurements (Sections 4 to 5). The physics and analysis parameters used throughout are the FM1 defaults of `het_ifo_opd.config.OPDConfig`.

2 Measurement principle

2.1 Frequency modulation encodes the OPD

A two-beam interferometer with optical path-length difference OPD converts the instantaneous optical frequency $\nu(t)$ of the laser into an interferometric phase

$$\varphi(t) = \nu(t)\,\frac{\mathrm{OPD}}{c} \qquad \text{[cycles]}, \tag{1}$$

equivalently $\varphi = \mathrm{OPD}/\lambda$. Modulating the laser frequency by a small amount $\delta\nu(t)$ about its mean therefore produces a phase modulation

$$\delta\varphi(t) = \frac{\mathrm{OPD}}{c}\,\delta\nu(t), \tag{2}$$

because $\mathrm{d}\varphi/\mathrm{d}\nu = \mathrm{OPD}/c$. The relation is linear: every Fourier component of $\delta\nu$ maps to the same component of $\delta\varphi$, scaled by the constant OPD/c. Driving the laser-frequency actuator with a single tone $\delta\nu(t) = \delta\nu_{\mathrm{pk}}\cos(2\pi f_{\mathrm{mod}}t)$ thus imprints a phase tone of amplitude

$$A_\varphi = \frac{\mathrm{OPD}}{c}\,\delta\nu_{\mathrm{pk}} \qquad \Longrightarrow \qquad \boxed{\mathrm{OPD} = \frac{A_\varphi\, c}{\delta\nu_{\mathrm{pk}}}} \tag{3}$$

where the peak (zero-to-peak) frequency deviation follows from the known drive and actuator gain,

$$\delta\nu_{\mathrm{pk}} = \frac{\mathrm{d}\nu}{\mathrm{d}V}\cdot\frac{V_{\mathrm{pp}}}{2} = 376\,\mathrm{MHz\,V^{-1}} \times \frac{1.0\,\mathrm{V_{pp}}}{2} = 188\,\mathrm{MHz}. \tag{4}$$

Two features make this attractive. It is wavelength independent: the OPD comes from the frequency modulation alone, and the 1550 nm carrier enters only as context. And it needs only the amplitude of a single tone, so the large, slowly drifting DC interferometer phase is irrelevant. The whole problem reduces to measuring A_φ at f_{mod} as accurately as possible.

2.2 Why the differential phase

The reference and measurement interferometers of a system share the same laser, so each channel is dominated by the laser frequency noise coupling through that interferometer's own absolute OPD, the tens-of-cycles wander noted in Section 1.2. Their difference $\varphi = \varphi_{\mathrm{meas}} - \varphi_{\mathrm{ref}}$ cancels this common term and leaves the residual differential OPD, which for a released system carries the test-mass motion. Figure 1 shows this for the clean anchored acquisition (Case A): the two channels wander by $\sigma \approx 4.31$ cyc each, while the difference has $\sigma_{A-B} \approx 0.0071$ cyc, a common-mode rejection of $\sim 606\times$. The injected tone is common to both channels as well and is likewise strongly rejected; what survives in the difference is the tone due to the residual OPD mismatch. Quantifying the tone in each channel gives absolute OPDs of 45.820 and 46.892 mm for the two interferometers, versus 1.072 mm in the difference. The package therefore uses the differential phase $\varphi = \mathrm{ch1} - \mathrm{ch2}$ as the observable and retains the two absolute OPDs as diagnostics.

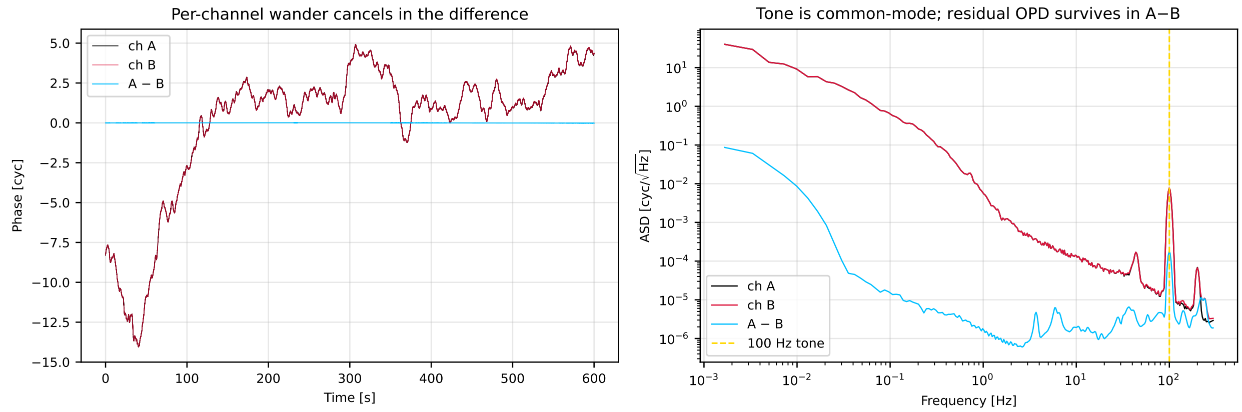

Figure 1: Common-mode rejection in the clean anchored acquisition (Case A). Left: the two interferometer channels (blue, orange) wander by several cycles, driven by laser frequency noise through their absolute OPDs; the difference (green) is smaller by $\sim 606\times$. Right: amplitude spectral density. The 100 Hz modulation tone (dashed) is large in each channel and reduced, but cleanly measurable, in the difference. The differential phase is the observable.

3 The estimation pipeline

The task of Section 2, measure A_φ at f_{mod}, looks elementary but hides a trap that dominated the early analysis. This section states the trap and the pipeline that removes it. The implementation lives in `het_ifo_opd.estimators` and `het_ifo_opd.pipeline`.

3.1 The leakage trap of a naive lock-in

The differential phase is still dominated by a large low-frequency wander (residual laser-noise coupling, plus real slow test-mass motion on released systems). A rectangular-window least-squares fit, or a plain DFT, at f_{mod} has spectral sidelobes that fall off only as $1/f$, so this enormous out-of-band content leaks into the f_{mod} bin and biases the amplitude high. Because the leakage scales as $\sim 1/T$, a full-record lock-in can be biased by up to $\sim 2\times$ and short per-segment fits by $\sim 10\times$: this was the origin of the long-standing puzzle in which the per-segment OPD sat far above the full-record number. A polynomial detrend and a windowed fit help, but the clean fix is to isolate a narrow band around the tone before measuring it.

3.2 Primary estimator: complex-baseband demodulation

We heterodyne the tone to DC and low-pass to a narrow one-sided bandwidth B:

$$z(t) = 2\,\mathrm{LPF}_B\big[\,\varphi(t)\,e^{-2\pi i f_{\text{mod}} t}\,\big], \qquad |z(t)| = A_\varphi(t) \quad \text{[cyc, zero-to-peak]}. \tag{5}$$

The low-pass rejects everything outside $[f_{\text{mod}} - B,\, f_{\text{mod}} + B]$, so the out-of-band wander physically cannot leak in. This makes $|z(t)|$ the instantaneous tone amplitude, hence the instantaneous $\mathrm{OPD}(t) = |z(t)|\,c/\delta\nu_{\text{pk}}$, while $\arg z(t)$ is the true tone phase, whose slope is any residual drive or digitizer frequency offset. The default bandwidth is $B = 0.5\,\text{Hz}$: wide enough to pass real tone dynamics, narrow enough to reject the broadband noise and suppress leakage. The FIR low-pass uses a Kaiser window for a deep, controlled stopband, and the tone frequency is taken from the known drive and refined on a fine periodogram grid (`estimators.refine_frequency`).

3.3 Coherent versus incoherent integration

From $z(t)$ we form a coherence

$$\rho = \frac{|\langle z \rangle|}{\sqrt{\langle |z|^2 \rangle}} \in [0, \sim 1], \tag{6}$$

after removing a constant frequency offset, and let it choose how to collapse $z(t)$ to a single number:

- $\rho \geq 0.7$ (`coherence_threshold`): the tone is phase-coherent over the record; report the coherent amplitude $|\langle z \rangle|$, the optimal-SNR estimate.
- $\rho < 0.7$: the tone phase wanders (the OPD is physically moving); report the noise-debiased incoherent amplitude $\sqrt{\langle |z|^2 \rangle - \langle |z_{\text{off}}|^2 \rangle}$, the correct estimator for a genuinely non-stationary tone, where z_{off} is an off-tone demodulation measuring the in-band noise.

This mechanism keeps a moving-test-mass acquisition from being reported as a falsely precise single value; for such records the instantaneous $\text{OPD}(t)$ is the real deliverable and the single number is a summary.

3.4 Uncertainty: noise floor and physical motion

Two distinct quantities are reported, and conflating them is itself a pitfall. The coherent noise floor is the statistical measurement limit,

$$\sigma_{A_\varphi} = \sqrt{\frac{\langle |z_{\text{off}}|^2 \rangle}{N_{\text{eff}}}}, \qquad N_{\text{eff}} = T \cdot 2B, \tag{7}$$

with N_{eff} the number of independent looks. This equals the Cramér-Rao bound $\sqrt{S_n(f_{\text{mod}})/T}$ for the local one-sided noise PSD S_n, and is verified calibrated against Monte-Carlo in the package tests. It shrinks as $1/\sqrt{T}$ only while the tone is coherent. The physical drift is the standard deviation of $\text{OPD}(t)$ over the record, a real motion of the test mass rather than a measurement error, and it does not shrink with T. On anchored records the noise floor dominates and the OPD is a precise number; on released records the physical motion dominates by orders of magnitude and the OPD is a moving target, which is exactly the force-sensing signal the released configuration is built to capture.

3.5 Cross-checks and quality diagnostics

The demodulator is cross-checked and flagged by three further estimators and ratios, all reported per acquisition:

- `leakage_ratio` $= A_{\text{lockin}}/A_{\text{demod}}$: the rectangular-window lock-in is retained purely as a leakage diagnostic. A value near 1 means the record is clean; a value much greater than 1 means the low-frequency wander is leaking into a naive estimate.
- `method_agreement` $= |A_{\text{demod}} - A_{\text{spec}}|/A_{\text{demod}}$: the demodulator against an independent windowed single-bin LPSD estimate (`speckit`). Both are leakage-suppressed, so a large disagreement now flags a real problem, typically incoherence, rather than leakage.
- `harmonic_ratio`: the second-harmonic amplitude relative to the fundamental, a live gauge of drive or actuator nonlinearity.
- `tone_snr` $= A_{\text{demod}}/\sigma_{A_\varphi}$: the coherent amplitude signal-to-noise ratio.

4 Three test cases

We now run the pipeline on three FM1 acquisitions chosen to span its operating regimes. All three use the FM1 physics defaults; the two Day09 records use $f_{\mathrm{mod}} = 95\,\mathrm{Hz}$ and the Day06 record 100 Hz. Cases A and B are the same system (R1), anchored and released respectively; Case C is the other system (R2) under the same released conditions as Case B. Both released cases are shimmed and flight-torqued in vacuum. Table 1 collects the headline results and diagnostics.

Case	Configuration	OPD [mm]	noise [µm]	drift [µm]	mode	ρ	leak $\times$	SNR
A	R1, anchored, air	1.0714	0.41	20.0	coherent	1.000	1.00	2611
B	R1, released, vacuum	0.0892	1.42	24.6	coherent	0.996	1.84	63
C	R2, released, vacuum	0.2713	9.79	158.3	incoherent	0.090	1.03	28

Table 1: Headline results for the three test cases. Case A is the clean, anchored reference. Case B is coherent but strongly leakage-biased (the naive lock-in over-reads by $\times 1.84$). Case C is genuinely phase-incoherent ($\rho = 0.090$), motion-limited, and correctly reported with the incoherent estimator. "noise" is the coherent noise floor (Eq. 7); "drift" is the standard deviation of OPD(t).

4.1 Case A: clean anchored reference

`FM1Day06_AnchoredAirOPD_EDUbase_R1_20260609_135823.zip`. This is system R1 with its test mass anchored, the best-behaved regime and the calibration point for reading the diagnostics. The tone sits far above the noise (SNR $\approx$ 2611), the demodulator, the windowed single-bin and even the rectangular lock-in all agree (`leakage_ratio` $= 1.00$, `method_agreement` $= 2.7e-04$), and the tone is fully coherent ($\rho = 1.000$, mode `coherent`). The residual OPD is 1.0714 mm with a coherent noise floor of only 0.41 µm; the 20.0 µm of drift in OPD(t) is small residual motion, not a limitation. The second-harmonic content is modest (`harmonic_ratio` $= 1.29e-02$). Figure 2 shows the three-panel diagnostic: the ASD with the tone marked, the phase-folded (synchronous-averaged) tone, and the flat OPD(t).

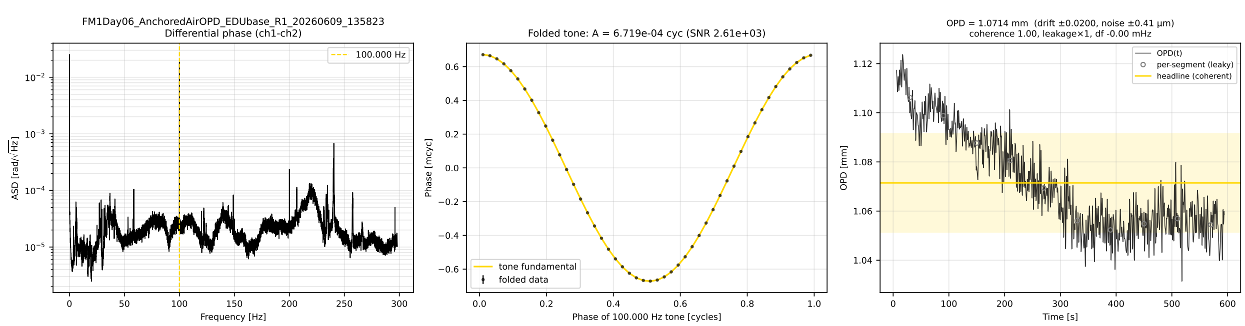

Figure 2: Case A diagnostics (R1, anchored, air). Left: differential-phase ASD, tone at 100 Hz marked. Centre: the phase-folded tone stands cleanly above the noise. Right: OPD(t) is essentially flat; the headline, drift band, and (here coincident) per-segment estimates all agree. This is the regime in which the frequency-modulation method delivers a precise static OPD.

4.2 Case B: released, coherent but leakage-biased

`FM1Day09_VacuumReleasedOPD_SHIMMED_FLIGHTTORQUED_R1_20260629_140109.zip`. This is the same system R1 as Case A, now with the test mass released and the unit shimmed and flight-torqued in vacuum. Over this particular record the tone remains phase-coherent ($\rho = 0.996$, mode `coherent`), and the demodulator returns a well-defined 0.0892 mm. But the low-frequency wander is now large enough that a naive rectangular lock-in over-reads by $\times 1.84$ (it would report 0.1638 mm), while the leakage-immune demodulator and the windowed single-bin agree far better (`method_agreement` $= 3.6e-02$; single-bin 0.0924 mm). This is the textbook leakage case of Section 3.1, caught by `leakage_ratio`. Note also the large second harmonic (`harmonic_ratio` $= 3.35e-01$): under torque the drive or actuator chain is visibly nonlinear, a point we return to in Section 5. Because the demodulator reads only the fundamental in a narrow band, that distortion does not corrupt the OPD, but it is a warning flag. Figure 3 shows the diagnostic; the leakage is visible as the per-segment markers sitting above the demodulated OPD(t).

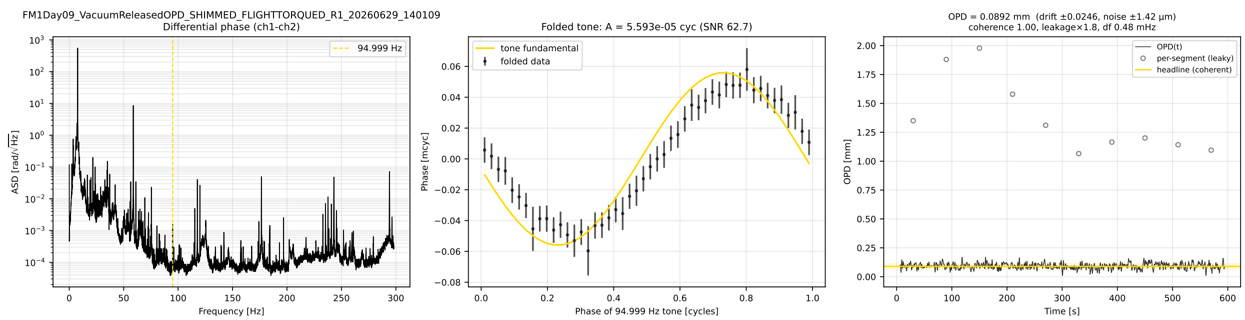

Figure 3: Case B diagnostics (R1, released, vacuum). The tone is coherent, so OPD(t) is a well-defined 0.0892 mm, but the low-frequency wander biases the rectangular per-segment estimates (grey open markers) high, the same leakage that inflates the naive lock-in by $\times 1.84$. The demodulator is immune to it.

4.3 Case C: released, phase-incoherent (motion-limited)

`FM1Day09_VacuumReleasedOPD_SHIMMED_FLIGHTTORQUED_R2_20260629_140110.zip`. This is the other system, R2, under the same released, shimmed, flight-torqued, vacuum conditions as Case B, and it behaves very differently. The test mass is moving over the record, so the tone is not phase-coherent ($\rho = 0.090$, well below 0.7) and the pipeline correctly falls back to the incoherent estimator (mode `incoherent`). The reported OPD is 0.2713 mm, but the meaningful quantity here is the motion: the drift of OPD(t) is 158.3 µm, more than an order of magnitude above the 9.79 µm noise floor. The leakage ratio is now near unity (`leakage_ratio` $= 1.03$), so this record is not leakage-biased, yet `method_agreement` is comparatively poor (5.9e-02), the expected signature of a genuinely non-stationary tone rather than of leakage. That the two systems have different absolute OPDs (about 46.2 mm for R1 in Case B versus 69.5 mm for R2 here) confirms they are distinct setups, not repeats. Figure 4 shows the wandering OPD(t): here the time series and its statistics, not a single value, are the deliverable.

4.4 Cross-case comparison

Figures 5 and 6 summarise the three cases side by side. Figure 5 plots the three amplitude estimators as OPDs: for the clean Case A they coincide, for Case B the rectangular lock-in stands well above

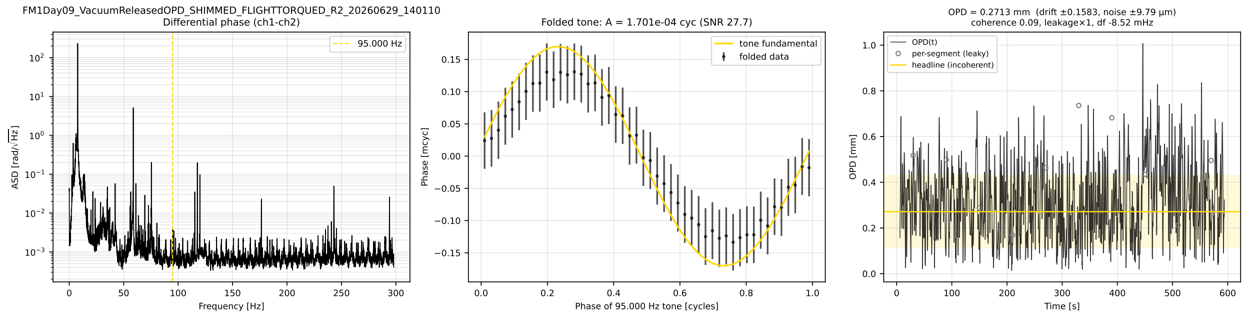

Figure 4: Case C diagnostics (R2, released, vacuum). OPD(t) moves by $\sim 158.3\,\mu\mathrm{m}$ across the record; the tone loses phase coherence ($\rho = 0.090$), so the incoherent estimator is used. For a released test mass this motion is the science signal, and the deliverable is the time series and its statistics, not a single precise value.

the two leakage-immune estimators, and for Case C all three are close (no leakage) even though the record is the least meaningful as a single number. Figure 6 overlays the instantaneous OPD(t): flat and static (A), offset but stable over this record (B), and wandering with the released test mass (C). Two lessons stand out. First, the comparison of Cases A and B, the same system R1 anchored then released, shows how anchoring produces a static, precisely measurable OPD while releasing introduces both real motion and the low-frequency wander that leakage feeds on. Second, the comparison of Cases B and C, two different systems under identical conditions, shows that `leakage_ratio` and ρ are independent diagnostics: a record can fail either test on its own.

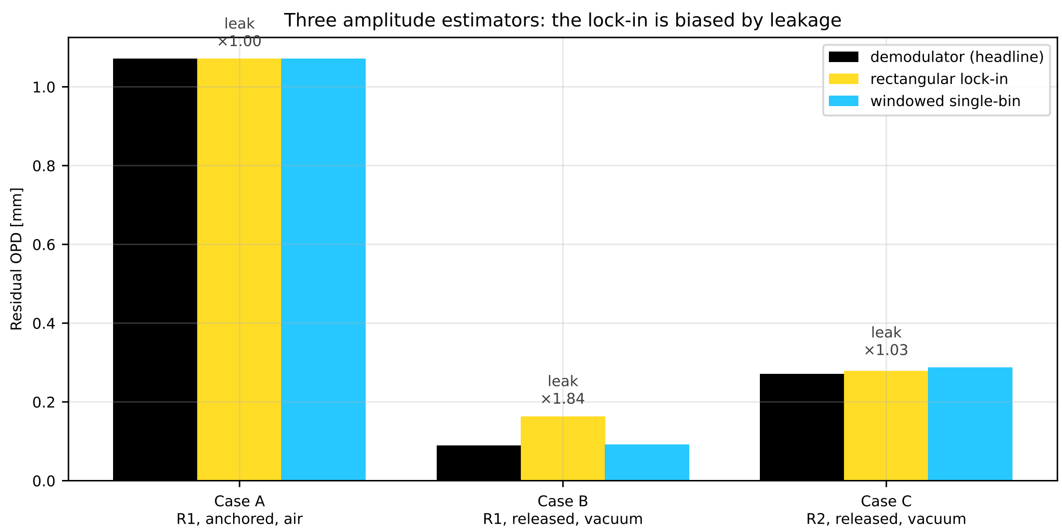

Figure 5: The three amplitude estimators, expressed as OPD, for each case. They agree when the record is clean (A), diverge when leakage is present (B, the lock-in over-reads by ×1.84), and agree again when the problem is motion rather than leakage (C). The rectangular lock-in is kept only as this diagnostic.

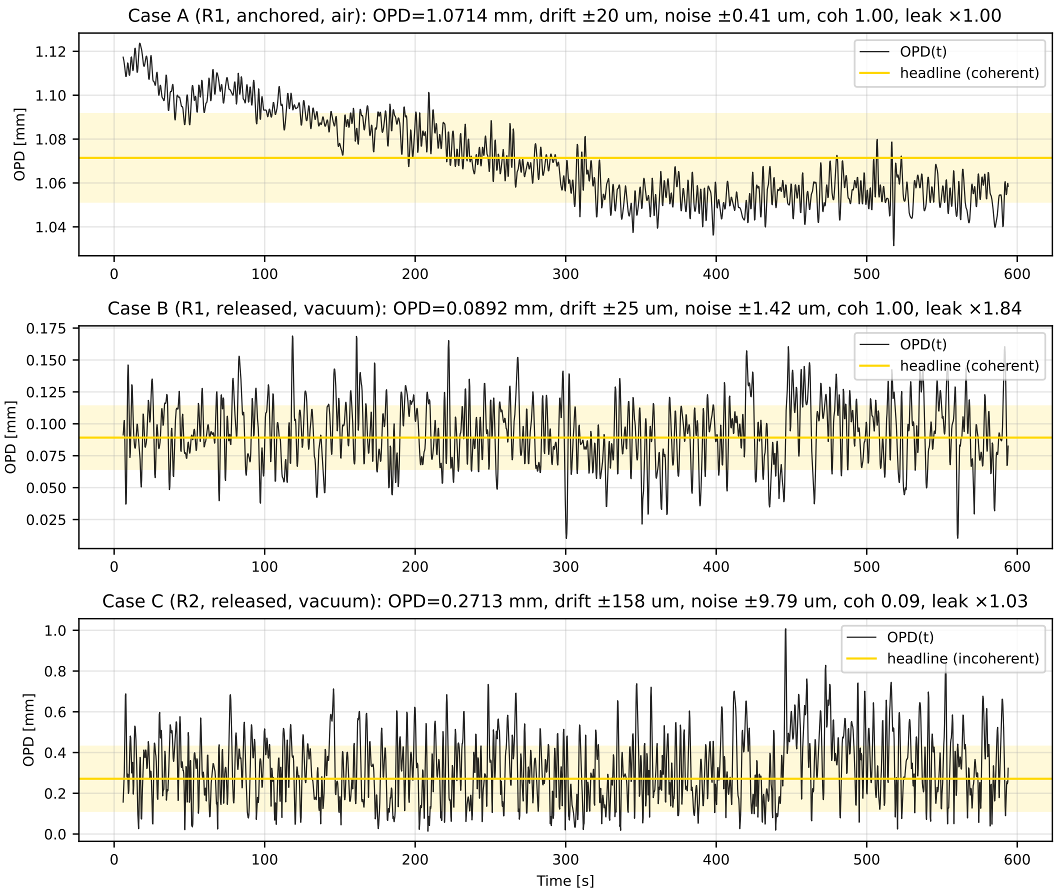

Figure 6: Instantaneous OPD(t) from the demodulator for the three cases (note the differing vertical scales). Case A is flat and precise; Case B is stable over this record but offset (and, before correction, leakage-biased); Case C physically wanders with the released test mass and is reported as a time series plus its motion statistics.

5 Pitfalls to avoid

The three cases, together with the wider FM1 set, make the recurring failure modes concrete. We list them roughly in order of how often they bite.

1. Spectral leakage from the low-frequency wander (Case B). A rectangular-window lock-in or plain DFT at f_{mod} leaks the tens-of-cycles low-frequency content into the tone bin and over-reports the OPD, by up to $\sim 2\times$ on a full record and up to $\sim 10\times$ on short segments. This is the single most damaging analysis error, and it is entirely avoidable: demodulate in a narrow band (Section 3.2) and treat any `leakage_ratio` much greater than 1 as a red flag. Do not quote short-segment rectangular estimates as the OPD.

2. Confusing real motion with measurement precision (Case C). When the test mass is released the OPD is time-varying by construction, so the coherent noise floor (Eq. 7) is not the uncertainty on "the" OPD; there is no single OPD. Reporting 9.79 µm for Case C would be off by more than an order of magnitude from the 158.3 µm of real motion. Watch the `mode` and ρ flags: a low coherence means report OPD(t) and its statistics, not a precise number. Longer records do not shrink the motion; they only characterise it better.

3. A released test mass has no single OPD. Following from the previous point: to obtain one static differential-OPD value the test mass must be anchored (Cases A). A released system is designed to let the mass move, and the demodulated OPD(t) is then the force-sensing signal. Do not force a single number onto a released record; report the trajectory.

4. Actuator or drive nonlinearity. The torqued Case B shows a large second harmonic (`harmonic_ratio` $= 3.35e-01$, against $1.29e-02$ for the clean Case A). Harmonics themselves do not corrupt the OPD, since only the fundamental is read, but they signal that the actuator may be compressing the fundamental, which biases the calibration $\delta\nu_{\mathrm{pk}} = (\mathrm{d}\nu/\mathrm{d}V)V_{\mathrm{pp}}/2$ low. Push the modulation amplitude up for SNR, but monitor `harmonic_ratio`, back off if it rises sharply, and calibrate $\delta\nu_{\mathrm{pk}}$ at the actual operating depth rather than trusting the small-signal $\mathrm{d}\nu/\mathrm{d}V$.

5. Calibration of the frequency deviation. $\delta\nu_{\mathrm{pk}}$ is a direct multiplicative scale on the OPD (Eq. 3), yet it is currently assumed from a nominal $\mathrm{d}\nu/\mathrm{d}V = 376\,\mathrm{MHz\,V^{-1}}$ taken as flat. Any error in $\mathrm{d}\nu/\mathrm{d}V$, in its frequency dependence at f_{mod}, or in the actually delivered V_{pp} maps one-to-one onto the OPD. This is the leading accuracy systematic and deserves a dedicated in-situ calibration.

6. Modulation-frequency choice and clocking. Place f_{mod} in a quiet valley of the differential-phase noise, away from line harmonics and mechanical resonances, and keep it and its first few harmonics comfortably below Nyquist. Share a common clock between the drive source and the phasemeter: an unshared clock leaves a residual tone-frequency offset (for example $-8.52\,\mathrm{mHz}$ for Case C) that the pipeline must fit out and that can erode coherence over long records.

6 Summary and recommendations

A known single-tone laser-frequency modulation turns each interferometer pair into a self-calibrating OPD sensor: the residual OPD is $\mathrm{OPD} = A_\varphi c/\delta\nu_{\mathrm{pk}}$, read from the amplitude of the tone in the differential phase, wavelength independent and immune to the large DC interferometer phase. The estimation pipeline is a complex-baseband demodulator that is immune to the spectral leakage which biased the naive lock-in, yields the instantaneous OPD(t), chooses coherent versus incoherent integration from a measured coherence, and reports an uncertainty equal to the Cramér-Rao bound. Applied to three FM1 acquisitions it recovers a sub-micron-precise 1.0714 mm on the clean anchored record (R1), correctly de-biases the leaky released record (R1) from 0.1638 mm (lock-in) to 0.0892 mm, and correctly reports the incoherent released record (R2) as a moving OPD(t) rather than a false precise value.

For the experimentalist, the levers that most improve a static differential-OPD measurement are, in order: anchor and mechanically quiet the interferometer (this gives a well-defined static OPD and enables long coherent integration); increase the modulation amplitude for SNR while monitoring the second harmonic (pitfall 4); calibrate $\delta\nu_{\mathrm{pk}}$ in situ (pitfall 5); choose f_{mod} in a noise valley with a shared clock (pitfall 6). For the analyst, the rules are: never quote a rectangular short-segment amplitude as the OPD (pitfall 1), and always read the `leakage_ratio`, coherence and `mode`, and `method_agreement` flags before trusting a single number. When coherence is low the mass is moving, so report the OPD(t) time series and its statistics instead.

Reproducibility

Every number and figure in this note is produced from the raw FM1 files by `make_figures.py`, which runs `het_ifo_opd.pipeline.estimate_opd` on the three cases, writes the figures to `Figs/`, and emits the numeric macros in `values.tex`. Rebuild the whole note with `make` (which runs the script and then `pdflatex`), or regenerate only the figures and values with `make figs`.

References

- Package and pipeline: `het_ifo_opd` (`estimators.py`, `pipeline.py`, `physics.py`, `config.py`); the end-to-end walk-through in `notebooks/fm1_opd_analysis.ipynb`; project `README.md`.
- Estimator validation (leakage-immunity, calibrated uncertainty near the Cramér-Rao bound, incoherent fallback): `tests/`.
- Spectral tooling: `speckit` (LPSD single-bin and windowed ASD); Moku:Pro Phasemeter I/O: `mokutools`.